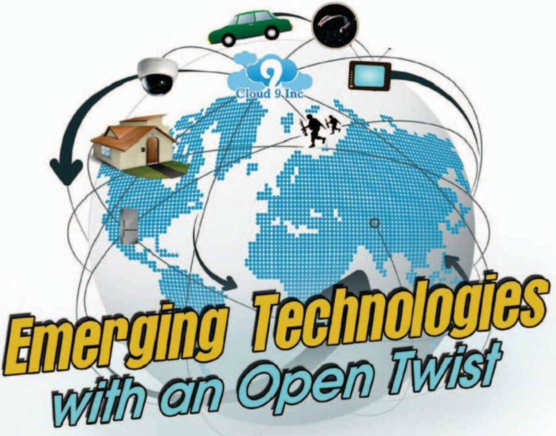

Emerging Technologies with an Open Twist

Open source is accelerating the growth of many emerging technologies—ranging from cloud management to defence infrastructure. Here are five such tech spaces to follow, and contribute to, in 2014. Watch out for five more in the next issue.

With open source, it becomes possible for everyone to learn, use, customise, improve and bring emerging technologies into mainstream use. Whether you do research on drug discovery, create animated content to teach complex concepts, build an automatic water spray for your garden, develop an Android-based mobile system to bring together students in your school, or simply customise and use a collection of apps to keep track of and improve your health, open source gives you an opportunity to be an inventor and innovator. It is therefore important for open source contributors to be aware of what's hot and what's not at any point of time, so that they can play a part in bringing emerging technologies to the fore.

On Cloud 9, with open source cloud management

Trend: The story started a long time ago with software-as-a-service (SaaS), and widened in scope to infrastructure-as-a-service (IaaS), platform-as-a-service (PaaS), and now anything-as-a-service (XaaS)! From data and software to whole IT set-ups, companies and individuals are using tech and tools over the cloud today—on clusters of shared computing resources owned and managed by cloud service providers. Having everything on the cloud is great because it can be accessed from anywhere using the Internet. But, it does pose a huge challenge too—will your resources get locked into a format that you cannot access later, will you lose the flexibility of modifying your software to suit your needs, will you lose the luxury of combining your specialised tools with the service provider's platforms? Open source cloud management lays such worries to rest.

Trend+: IT pros are quite excited about open source cloud management these days, and there are many great solutions such as OpenStack, Apache's CloudStack and OW2. While the first two focus on IaaS, OW2 focuses on PaaS.

CloudStack offers a whole stack of features for IaaS. These include compute orchestration, network-as-a-service, user and account management, a full and open native applications programming interface (API) and resource accounting—all with a very intuitive user interface (UI) too. It supports VMware, KVM, XenServer and Xen Cloud Platform (XCP), and the API is compatible with AWS EC2 and S3, enabling organisations to deploy hybrid clouds with ease.

Founded by NASA, and supported by industry leaders like Rackspace, Ubuntu, Nebula, Red Hat and IBM, OpenStack